

An Analysis of the Long-Tail Impacts of Subprime Loans in the Early 2000s

Matthew Robson

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1 Introduction

1.1 Subprime Loans

Despite the wealth of knowledge around the intimate impacts of subprime loans on the 2008 financial crisis, there is little known about how these loans may have impacted the housing market longer term. The prevalence of subprime loans began in the early 2000s and persisted up to the financial crisis of 2008. These subprime loans were given to high risk borrowers. Being so, these loans carried higher interest rates or stricter terms, such as larger down payment requirements or steeper penalties. These predatory loans originated in adjustable-rate mortgages—mortgages where the interest rate for the borrower may change based on external economic circumstances [1]. Adjustable-rate mortgages often lead to an exploitative structure due to the increased rates over time. This coincides with subprime loans because both generally have harsher terms and so are taken out by people who have poor credit. This increase in subprime loans during the early 2000s may have driven a shift toward single-family residential construction relative to multi-unit housing. This is because many people who previously did not have access to taking out a loan for the construction of their own house now were given the opportunity as more loan agencies were offering subprime loans to a much wider set of people.

1.2 The 2008 Financial Crisis

Subprime loans like these were then traded on the public market in many different forms such as collateralized debt obligations (CDOs) and mortgage-backed securities (MBS). A CDO is a financial device that consists of many mortgages and spreads the risk into "tranches" with differing return rates and risk rates. Generally, the rating scheme for CDOs is considered highly misleading because while the senior tranches might be more secure than the mezzanine or equity, the senior tranche is still backed by loans that may default, leading to loss of profits. MBS are nearly equivalent, however, where CDOs target all types of loans, MBS are strictly housing backed loans.

The 2008 financial crisis is often attributed to the over-reliance on subprime loans and over-leveraging of the early 2000s. Because many firms in the early 2000s acquired high-risk subprime assets, when those borrowers eventually defaulted on their loans, the firms did not have the capital to back up their highly leveraged positions and, in some cases, were forced to be fully dissolved [5]. The consequences of this crisis have been seen across both the housing and economic sectors. In the housing sector, the 2008 crisis led to a crash in housing prices and the fire sale of a large portion of properties.

1.3 The Current Housing Crisis

The current housing market is full of difficulty for new and low-income buyers. Average house prices have been rising relative to average income for the past few years, leading to a high cost burden on those wanting to purchase a house. Specifically, we have seen an increase of 400% in median housing prices in the past 40 years, with increase of only 300% in median income [10]. In some cases, nearly 70% of a household's income is used directly to pay rent [9]. In addition to this, housing has become so overcrowded that many people are forced to live in cramped conditions [8]. Proposed causes for this crisis such as the over-indebtedness of households today and rapid urbanization outpacing housing development appear to clarify this connection, but these claims will not be rigorously evaluated in this study.

1.4 Theoretical Impacts of Subprime Loans

The expansion of subprime loans in the early 2000s plausibly affected long-run housing outcomes such as with housing insecurity in the late 2010s. The first way in which housing insecurity may be impacted is through overcrowding. As greater proportions of housing were zoned for single-family housing rather than multi-family housing and zoning is very sticky in the US context, we may continue to see impacts of less accessible housing for the decades to come. This would provide us a positive relationship between subprime percentage and overcrowding in homes. The second factor that may be considered for housing insecurity is cost burden. As subprime loans led to increased interest rates for loans, the long term impacts of these rates may have lead to a perpetuation of high interest rates and difficulty in purchasing housing in certain areas.

2 Methods

2.1 Analysis Framework

In order to preform this analysis we leveraged python and its associated packages. Specifically: NumPy, Polars, and Pandas were used to manipulate and structure the data from the given sources. In order to perform the analysis, Matplotlib, StatsModels, and GeoPandas were used. We used the included

OLS methods from StatsModels to perform the bitartrate and multivariate regressions for both the overcrowding and cost burden models. For the purposes of this paper, we will consider any loan with a rate spread greater than or equal to 3.00% as subprime. This has some drawbacks in cases where the rate spread may not indicate some other requirements of the contract or the loan or may be measured inaccurately [7].

2.2 Data

We collected loan data for this study from the HMDA, focusing on loans originated between 2004 and 2007. Data from 2000 to 2006 was sourced from OPENICPSR [3], and data from 2007 – 2010 was sourced from the Consumer Financial Protection Bureau website [2]. These data were used to generate the percentage of subprime loans for each census tract over this time period. Specifically, we focused on the rate spread variable to classify loans as subprime or prime. We used the ACS 5-year estimates [11] to obtain the data for our dependent and control variables. We used data from the Housing, Socioeconomic, and Demographic datasets for this analysis. In order to maintain normalized and well sectioned areas we constructed a cross-sectional dataset at the census tract level. Some of the models included use spatial data for the shapes of the census tracts, and this data came from IPUMS NHGIS [6]. While the dependent variable (housing insecurity) is being measured ten years after the measurement of the independent variable (2016 – 2020), we are explicitly looking at how subprime loans may have very long term impacts on housing.

2.3 Dependent Variables

The dependent variable considered for this study is housing insecurity. To operationalization housing insecurity, we look at cost burden and overcrowding rates. In this study, we consider someone to be "cost burdened" by their housing if more than 35% of their income has gone to housing in the past 12 months. Rather than the cutoff supported by the HUD (30%), we chose 35% as our cutoff to account for areas with high cost living and non-fixed housing costs. We consider the overcrowding rate of a census tract to be calculated with the following formula (Housing Units With More than 1 Person per Room) / (Number of Housing Units). Overcrowding will be considered to be a rate from [0-1] whereas cost burden is a scaled percentage between 0 and 100.

2.4 Lending Variable

The variable used for lending was rate spread. Rate spread is defined as the loan interest rate minus the average loan interest rate. We classified any value above 3% as subprime lending and any value below 3% as prime lending [2]. This variable is reported to HMDA by lending institutions. When banks give out loans, they store the information about the loan and report the loans by March 1st of the following year. This produces a comprehensive dataset of

every loan originated that year, along with location, income, demographic, and other associated data. From here we extracted the loans that exceed 3% in rate spread and classified those as subprime. We then found the percentage of subprime loans out of all loans taken out in a certain census tract and calculated the percentage of subprime loans for each census tract. This percentage is measured as a rate between 0 and 1.

2.5 Control Variables

In order to control for variability in our predicted variables, we included some control variables. We selected to include: White %, Black %, Hispanic %, Undergraduate Graduation % (of total population), Median Income, Unemployment Rate, and Poverty Rate. All of these variables are taken from the ACS (5yr) report from 2016 to 2020. These data come from surveys and estimates made by ACS over this five-year period [11]. All of these measured percentages are values are normalized between 0 and 1, however median income purely the median income of the given tract measured in thousands of USD.

3 Results

We ran our regressions on both dependent variables—overcrowding rate and cost burden. For the overcrowding rate regression, we found that the subprime loan rate was statistically significant in the baseline model and the model including economic variables. When including demographic variables, subprime loan rate lost significance. The subprime rate held its lack of significance after adding both the economic and demographic variables. Our best model for the overcrowding regression achieved an $R^2 \approx 0.404$.

On the other hand, in the cost burden regressions, we found that subprime loan rate was significant across all of our models. We saw a loss in predicting power after adding economic variables from 37 to 12. Additionally, our best model achieved an $R^2 \approx 0.298$ when predicting cost burden. This is significantly lower than that of the overcrowding model.

Table 1: Regression Results: Overcrowding Rate

	<i>Dependent Variable: Overcrowding Rate</i>			
	(1) Baseline	(2) Economic	(3) Demographic	(4) All Variables
Subprime Loan (%)	0.1562*** (0.0048)	0.1029*** (0.0065)	0.0062 (0.0074)	0.0024 (0.0073)
Unemployment Rate (%)		-0.0002** (0.0001)		-0.0008*** (0.0001)
Poverty Rate (%)		0.0009*** (0.0000)		0.0002*** (0.0000)
Median Household Income		0.0001*** (0.0000)		0.0001*** (0.0000)
White (%)			-0.0013*** (0.0000)	-0.0013*** (0.0000)
Black (%)			-0.0012*** (0.0000)	-0.0011*** (0.0000)
Hispanic (%)			-0.0002*** (0.0000)	-0.0002*** (0.0000)
Undergraduate (%)			-0.0004*** (0.0000)	-0.0005*** (0.0000)
Constant	0.0138*** (0.0005)	0.0039*** (0.0012)	0.1438*** (0.0028)	0.1425*** (0.0030)
Observations	44,611	44,489	44,611	44,489
R^2	0.0270	0.0581	0.3968	0.4044
Adjusted R^2	0.0270	0.0580	0.3967	0.4043

Standard errors in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Regression Results: Cost Burden

	<i>Dependent Variable: Cost Burden Rate</i>			
	(1) Baseline	(2) Economic	(3) Demographic	(4) All Variables
Subprime Loan (%)	-36.8989*** (1.0786)	-12.3028*** (1.4482)	-19.8165*** (1.7683)	-25.1691*** (1.6327)
Unemployment Rate (%)		0.2010*** (0.0158)		-0.0043 (0.0156)
Poverty Rate (%)		-0.0059 (0.0075)		-0.2112*** (0.0082)
Median Household Income		0.0786*** (0.0021)		0.0266*** (0.0022)
White (%)			-0.2518*** (0.0065)	-0.2754*** (0.0066)
Black (%)			-0.1558*** (0.0071)	-0.1247*** (0.0064)
Hispanic (%)			-0.0795*** (0.0078)	-0.0678*** (0.0067)
Undergraduate (%)			0.1269*** (0.0036)	0.0525*** (0.0039)
Constant	20.2390*** (0.1329)	10.9132*** (0.3008)	32.2269*** (0.7134)	37.7072*** (0.6408)
Observations	44,716	44,489	44,716	44,489
R^2	0.0291	0.0722	0.2389	0.2981
Adjusted R^2	0.0291	0.0721	0.2389	0.2980

Standard errors in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

4 Discussion

Our most notable findings were the loss of predicting power of the subprime rate when predicting cost burden after adding economic variables, and the loss of significance of the subprime rate when predicting overcrowding after adding demographic variables. The loss of predictive power and significance of subprime loan percentage when adding demographic variables to the overcrowding model is revealing. This indicates that a portion of the information encoded by race is shared by subprime percentage. Additionally, we observe that the coefficient on the subprime rate decreases after adding economic variables in the cost burden model. This suggests that the subprime rate also captures a broader systemic economic variable impacting housing security.

The most notable finding here is the negative correlation between the subprime rate and cost burden. This was contrary to our initial expectation; however, this may be due to omitted control variables. The significant reduction in correlation between subprime rate and cost burden after adding economic variables suggests that including additional economic controls could account for this negative relationship. Specifically, considering the economic status during the loan process such as with the included income data from the HMDA data set could explain some of this negative correlation [2]. Additionally, our models omit several potentially relevant control categories, such as demographic trends, political factors, and boundary changes in census tracts.

As mentioned in earlier sections, there is much more to housing insecurity than the cost burden and overcrowding that we were able to consider in this study. Variables such as homelessness, safety, and access to public facilities could all be modeled to get a comprehensive view of the impacts of subprime loans on the current housing crisis [4].

5 Conclusion

The findings of this study show that while subprime loans may have a significant impact on housing insecurity, further research will be needed to conclusively show the impact. Taken at face value, the study shows that census tracts with a previously greater percentage of subprime loans should expect to have lower cost burdens. However, the study also shows that census tracts with a previously greater percentage of subprime loans should expect to be more overcrowded. Together, these results appear contradictory, indicating that additional confounding variables ought to be considered.

This suggests that investment stemming from a financial bubble can have both negative and positive impacts on infrastructure. Extending this framework to modern contexts, we might expect results similar to those of a potential AI bubble, drawing parallels to the earlier dot-com bubble.

References

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A Graphical Models

A.1 Subprime Loan Spread (2006)

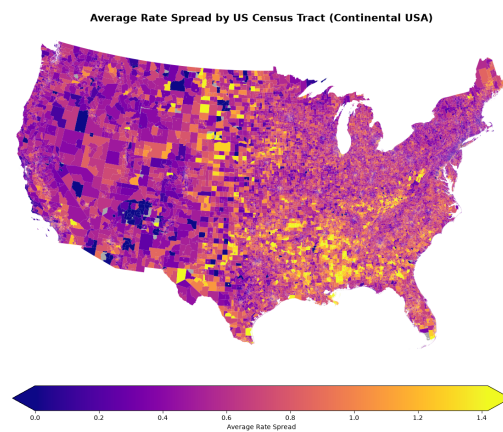


Figure 1: Geographic distribution of subprime loan rate spreads in 2006.

A.2 Variable Pairwise Relationships

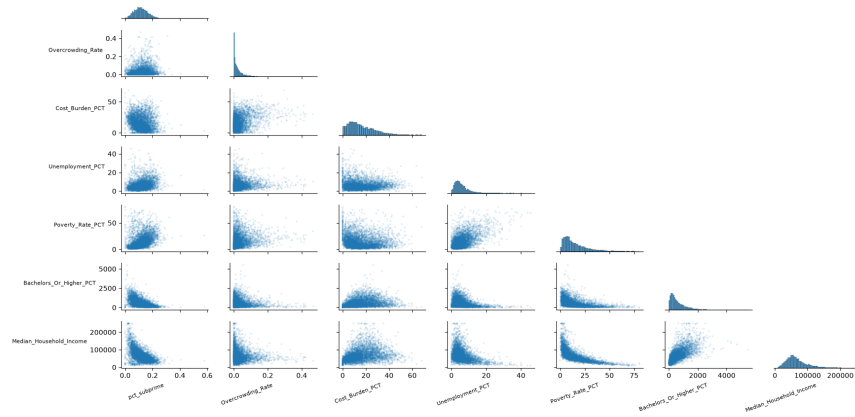


Figure 2: Pairwise relationships between variables used in regression analysis.